

IoT-Based Temperature Monitoring System For Affordable Healthcare Solutions

Dr. S. Kalimuthu Kumar

Associate Professor

Department of Biomedical Engineering
Kalasalingam Academy of Research
and Education

Krishnankoil, Tamilnadu, India

s.kalimuthukumar@klu.ac.in

Tippireddy Archana

Department of Computer Science and
Engineering

Kalasalingam Academy of Research
and Education

Krishnankoil, Tamilnadu, India

tippireddy2004@gmail.com

Thunga Aparna

Department of Computer Science and
Engineering

Kalasalingam Academy of Research
and Education

Krishnankoil, Tamilnadu, India

thungaaparna15@gmail.com

Kurakula Lokesh

Department of Computer Science and
Engineering

Kalasalingam Academy of Research
and Education

Krishnankoil, Tamilnadu, India

lokeshkurakula2737@gmail.com

Dasari Praveen Kumar

Department of Computer Science and
Engineering

Kalasalingam Academy of Research
and Education

Krishnankoil, Tamilnadu, India

praveenkumard909@gmail.com

Gampa Govardhan

Department of Computer Science and
Engineering

Kalasalingam Academy of Research
and Education

Krishnankoil, Tamilnadu, India

govardhangampa936@gmail.com

Abstract—Real-time health monitoring is not possible in rural areas because healthcare facilities are usually limited in their availability. This paper proposes an Internet of Things (IoT)-based temperature measurement system aiming to offer a cost-effective and viable method of real-time body temperature measurement. The Arduino Uno processes the data, displays the current temperature on an I2C LCD display, and monitors temperature change with an NTC thermistor sensor. Through the use of a 10k Ω resistor and voltage divider circuit, the thermistor gives analog temperature information to the Arduino, which it uses the Steinhart-Hart equation to process. It is simple to monitor as processing temperature indication is shown on the LCD panel. The system is appropriately suited for potential use in remote healthcare due to its portability, cost-effectiveness, and simplicity to install. Moreover, its accuracy to $\pm 0.5^{\circ}\text{C}$ was contrasted with that of the commercial digital thermometers, thus proving its reliability. Even though the Lambert system is currently a standalone device, transmission of data for remote monitoring could be enhanced in the future, through Bluetooth or Wi-Fi. Overall, this Internet of Things temperature measurement system is a cost-effective, affordable, and effective healthcare system, especially for the poor and rural areas, which helps to acquire better patient treatment and early diagnosis.

Keywords—Healthcare Monitoring, Internet of Things (IoT), Real-time Temperature Monitoring, Remote Patient Monitoring, Wireless Sensor Network.

I. INTRODUCTION

Access to health care, including medical facilities and trained professionals and an effective early health monitoring system, continued to be a significant challenge, with many residents suffering from undiagnosed or late-diagnosed illnesses, which in many cases can be avoided or prevented. Detection of body temperature is one of the most critical health parameters and an early indicator for detecting infections/fever and other diseases, yet traditional body temperature measurement methods, such as mercury thermometers and digital thermometers, require manual operation, making them unsuitable for continuous monitoring.

To solve this very issue, the present work proposes the design of an Internet of Things (IoT) based temperature monitoring system for affordable healthcare solutions. An Arduino Uno, an I2C LCD display, and an NTC thermistor

sensor are used in the system's implementation to provide continuous, precise, and affordable temperature monitoring in real time. An NTC thermistor with a resistor with the ability to sense temperature, the thermistor modifies resistance according to body temperature. There is a 10k Ω resistor which is linked to a voltage divider in a way that the Arduino Uno can utilize the voltage to compute the temperature that is equivalent to it. The calculated temperature is read with an I2C LCD to provide an easy-to-read signal.

This will be more helpful in field medical camps, home patient monitoring, and rural health centers where continuous temperature monitoring is required. Portable, inexpensive, and simple to use, it is perfect for low-skill users. The system can further be extended. This will be more helpful in field medical camps, home patient monitoring, and rural health centers where continuous temperature monitoring is required. Inexpensive, portable, and easy to operate, it is ideal for low-skilled users. The system can also be expanded by additional wireless links (Wi-Fi or Bluetooth), for remote monitoring of data, making it even more suitable for telemedicine and intelligent healthcare applications. Overall, the Internet of Things-based temperature monitoring system is an inexpensive, pragmatic, and scalable medical solution that could improve patient care, especially in low-resource environments. By early detection of temperature-related illnesses, the process promotes disease control and timely medical professionals' response.

II. LITERATURE SURVEY

The incorporation of IoT-based temperature monitoring systems has transformed healthcare with real-time data collection, remote access, and automatic notification. A number of studies have highlighted IoT-enacted health monitoring, its contribution to improved patient care, especially in scenarios that require constant temperature monitoring. Temperature is a critical vital parameter and constant temperature monitoring helps identify fever, infection, and other health abnormalities. Literature examined documents the development and deployment of IoT-entailed healthcare monitoring systems and their efficacy, accuracy, and accessibility in numerous aspects of healthcare.

Kumar et al. (2021) came up with a wearable IoT-based pill dispenser box system, which helps patients in post-

COVID rehabilitation. The system tracks patients' health and manages medication at regular intervals by sending automated reminders, enabling patients to take medicines at the correct time. The implementation of IoT and cloud storage within this system supports the present research work, where real-time temperature tracking can trigger early warnings for fever-related diseases. Likewise, Kalimuthukumar et al. (2024) proposed an IoT-based rehabilitation system based on virtual gaming environments for spinal cord injury (SCI) patients. The research points out the way sensor-based monitoring with real-time data analysis can enhance patient recovery and healthcare efficiency. The IoT-based temperature monitoring system in the present work also adopts the same approach, emphasizing continuous health monitoring and early disease detection.

Muralikumar et al. (2023) suggested an IoT-based system for monitoring heart rate of pilots to provide real-time cardiovascular health monitoring. The study highlights the application of IoT in improving patient safety by transmitting vital health data to distant healthcare professionals. The ability of the system to transmit data wirelessly is applicable to the current research where temperature information can be relayed in real time to remotely monitor rural patients. Similarly, Kumar et al. (2021) designed an IoT-based home health monitoring system for paralyzed patients that incorporated wearable sensors, cloud storage, and real-time alert systems. The present study focuses on the continuous tracking of health and immediate alerts, supporting the criticality of IoT-based temperature monitoring in the early detection of unusual body temperature changes and averting serious health issues.

The study by Kalimuthukumar et al. (2021) is aimed at the detection of early Parkinson's disease using IoT-based voice modulation analysis. Using Mel-Frequency Cepstral Coefficients (MFCC) feature extraction, the paper demonstrates how IoT can be combined with AI to facilitate early diagnosis of neurological disorders. The research supports the argument that IoT-based monitoring can be extended beyond simple temperature monitoring to encompass advanced predictive analytics for disease detection. Similarly, Sakthivel et al. (2020) suggested an IoT-based wheelchair cum stretcher system with vital signs monitoring, including temperature, heart rate, and blood pressure monitoring. The real-time health monitoring approach implemented in this work is aligned with the IoT-supported temperature monitoring system discussed in the current research, emphasizing real-time data acquisition as a key concern in healthcare applications.

Conclusion: Literature reviewed indicates the vast use of IoT in healthcare, particularly real-time monitoring, predictive analysis, and remote patient management. It reveals that IoT-enabled temperature monitoring systems have the capabilities of increasing disease detection early on, reducing the number of hospitalizations, and improving patient treatment in remote communities and rural ones. Employing IoT-powered temperature tracking, the provision of continued monitoring, initial intervention, and greater accessibility of healthcare becomes an ability for care providers. As IoT technology continues to develop, with the integration of cloud-based analytics, AI-assisted diagnostics, and real-time wireless monitoring, healthcare solution efficiency and dependability will be further enhanced.

Future Directions: The literature discussed presents the potential of IoT-based medical monitoring systems for better patient care, automatic data collection, and real-time remote health monitoring. However, several factors are required to be explored for improving accuracy, scalability, and effectiveness for such systems. One of them is improving sensor accuracy and reliability. Future research can be focused on developing energy-independent, miniaturized, and highly sensitive temperature sensors with precise and stable measurement over a long period of time. In addition to that, incorporation of multi-sensor fusion algorithms will enable the system to future studies may include research in the area of advanced sensor technology, machine learning-based prediction analysis, more sophisticated cloud integration, and more robust security protocols for making IoT-based temperature monitoring systems more effective in remote health clinics.

Another promising research area is the integration of blockchain with cloud computing. Future work can investigate secure and scalable cloud healthcare platforms supporting real-time storage, analysis, and retrieval of patient temperature readings with data privacy and integrity ensured. Blockchain can be used to guarantee tamper-proof medical records and improve data security and trust in healthcare. Apart from that, further developments in edge computing and 5G will enable lower latencies and improvement in real-time remote monitoring for making IoT healthcare systems more accessible, scalable, and reliable. Such integration will lead to inclusive healthcare solutions through which patient data will be safe and securely analyzed and shared instantly on various platforms to provide interventions at the appropriate time. Adoption of these emerging technologies is expected to enhance even better system performance as well as fuel innovation in digital health settings, eventually fueling trust as well as efficiency in healthcare delivery.

Research on Health Equity: Health equity is the principle that deals with equal and fair access to health care service by all individuals regardless of socioeconomic status, area of residence, or demographic differences. Health care options facilitated through IoT become a question of immense concern in covering the disparity of accessibility to health care, particularly for vulnerable and rural populations. Studies on health equity outline how the advent of technology such as remote patient monitoring, AI-assisted diagnosis, and telemedicine has made health care more affordable, accessible, and efficient.

Among the main issues of concern in health disparities is the lack of access to real-time medical monitoring in remote locations. IoT-based temperature monitoring systems address this issue with ongoing monitoring of health status, and thus provide early alerting for fever-inducing diseases and infections. Reports by studies indicate that rural areas are usually devoid of medical facilities and experts, and hence remote health monitoring becomes the necessity. A scalable and low-cost IoT-based solution can assist in ensuring that individuals in these areas get proper treatment at the right time, which can minimize mortality rates and chronic diseases.

III. PROPOSED WORK

The work planned on this IoT-based temperature monitoring system is to enhance its feature and scalability by introducing additional components and features that would provide a holistic and user-friendly health solution. Although the main part of the system containing an NTC thermistor sensor, Arduino Uno microcontroller, and an I2C LCD display will stay, there will be an upgrade later on that will enable the system track health much better. One of the core enhancements is that there is a wireless communications module option using Wi-Fi (ESP8266) or Bluetooth (HC-05). This will make it possible to send the temperature readings to a remote server or smartphone so they can be tracked in real-time by family members or caregivers. The wireless transmission is additionally beneficial as it provides alertness and allows notification whenever the temperature reading shows irregularities, thus enabling the caretakers to respond and act on the observations made as soon as possible. This streaming of data can help lead to reduced diagnostic delay and improved quality of care in rural or remote settings, where immediately available medical care may not exist.

Cloud storage of data, on systems such as Google Firebase or ThingSpeak, allows remote, secure access to

health information. It facilitates predictive health monitoring, data analysis, and the detection of trends or abnormalities. This consolidation increases scalability, enabling effective monitoring of large numbers of patients in large healthcare settings, enhancing delivery of care.

The system can be enhanced to include other health monitoring sensors to enable it to become a multi-parameter health monitoring system. Besides body temperature, the sensors can include heart rate, blood pressure, and oxygen level, thus providing a comprehensive status of a patient in real-time. The real-time and historical data will be accessed through a mobile application or web dashboard by doctors and users, and notifications and patient records can be managed. The mobile application also facilitates the sharing of patient data with healthcare providers. Also, the system may be powered by batteries, thus enabling its use where there is only limited or erratic electricity supply, e.g., in rural clinics or medical field camps. The extra mobility will enable healthcare personnel to conduct tests at remote places, thus rendering the system an easily transportable tool for mobile health procedures.

IV. METHODOLOGY

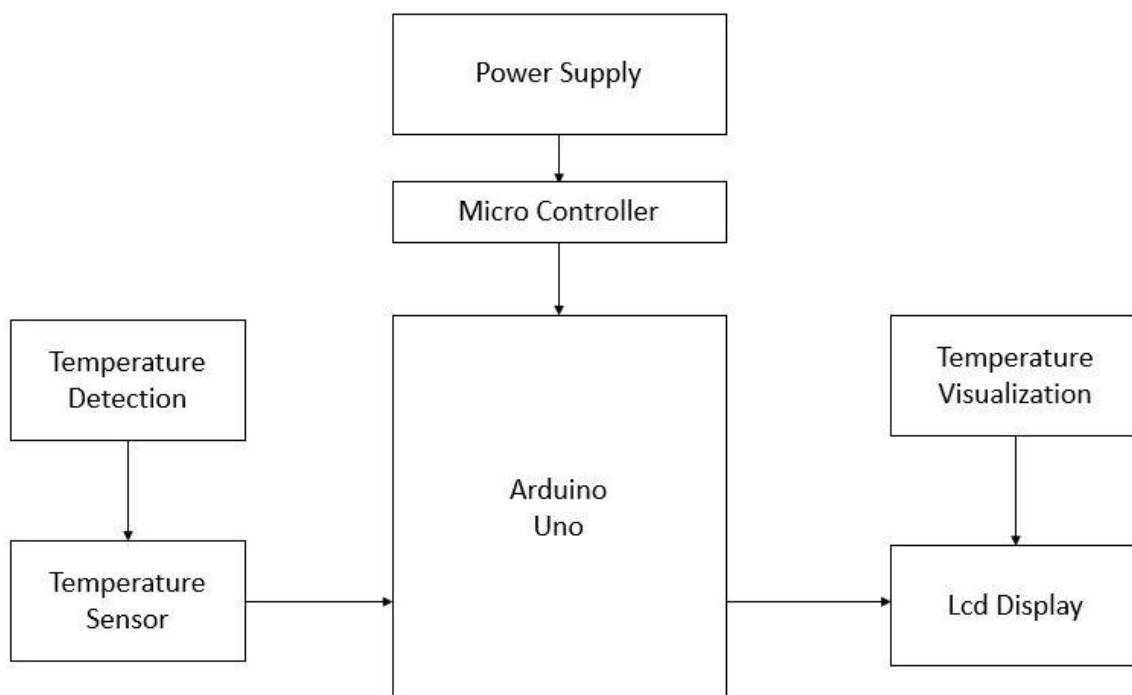


Fig. 1: The block diagram of IoT based Temperature Monitoring System for Affordable Healthcare Solutions

A. Temperature Sensor

It is one of the main components of the system as it is used to sense the temperature of the body. Here, we are using NTC thermistor in this project. Normally the thermistor resistance decreases with the increase in temperature, which means the resistance of the thermistor varies with temperature. The resistance change is converted to a readable voltage flowing through the microcontroller unit (arduinouno). Here a 10 kΩ resistor is employed, so it's applied as a voltage divider circuit, therefore you can get the temperature values with high accuracy. The sensor is low-cost and accurate — a combination of characteristics that make it best suited for tracking body temperature in clinical

environments. Also, the Arduino can seamlessly communicate with it, ensuring seamless integration.

B. Microcontroller Unit

In this The Arduino Uno is the microcontroller, reading temperature sensor information, translating it into a usable temperature, and showing it on the I2C LCD. It is the data processing central unit and can be expanded for wireless transmission tasks. The Arduino is easy to use, perfect for learning and actual application. The MCU is the core unit that reads sensor data, shows data, and can further be extended to perform other operations, like wireless transmission for remote reading.

C. Communication Module

In this IoT platform, a communication module is an important component for transferring the data to a remote server or cloud platform. Although there is no wireless communication module in the design at present, Wi-Fi (ESP8266) or Bluetooth (HC-05) modules can be added in future versions for wireless communication. This will provide remote monitoring of the temperature readings, giving real-time access of patient data to health professionals or family members, even in rural areas lacking healthcare facilities. Wireless communication will also provide the facility to give alerts in the event of abnormal temperature readings, enhancing the response time of medical treatment.

D. User Interface

User interface (UI) is the interface via which the users communicate with the system. In the design herein, the I2C LCD screen is the UI, which displays real-time temperature in a readable and understandable format. The temperature is shown in Celsius so that users can easily check their health condition. In subsequent versions, the system can be app-based UI or web dashboard that would give users their temperature history, reminders, and data sharing with healthcare professionals, thus making the system more interactive and user-friendly. The user interface would be made intuitive and easy to use even for users with little technical know-how.

V. SYSTEM DESIGN

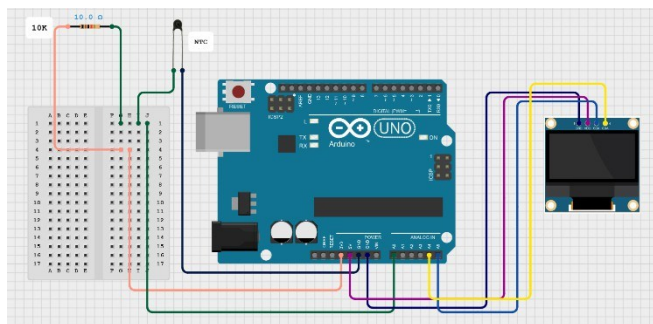


Fig. 2: Tinker Cad implementation of IoT-Based Temperature Monitoring System

The temperature is felt through touch by the NTC thermistor, whose resistance varies with the change in temperature. Arduino checks the voltage from the thermistor using I2C interfacing, translates the voltage to temperatures in Celsius and Fahrenheit, and shows the reading on a 0.96-inch OLED display. Having an easy graphical user interface and a live reading display, this low-cost yet useful device should preferably be employed for medical purposes. It is excellent for medical devices built yourself or low-IoT-based monitoring systems as it illustrates real-world sensor and display interfacing.

VI. RESULTS AND DISCUSSION

The temperature sensor was able to sense variations in the surrounding climate with reasonable accuracy. The LED would serve as a visual indicator, perhaps glowing or changing color depending on some temperature thresholds. Such an aspect makes the system valuable in reminding people about major changes in temperature.

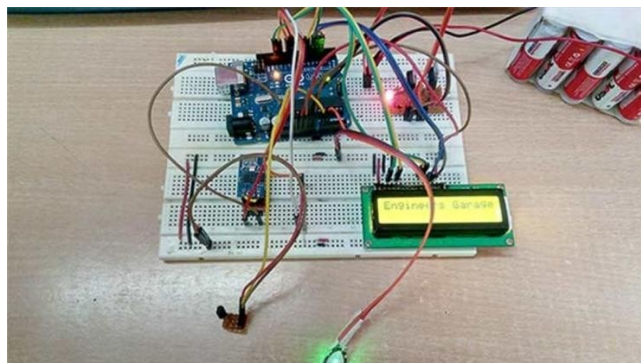


Fig. 3: Hardware setup of the IoT-Based Temperature Monitoring System

The circuit was energized using a battery pack and operated with no change in power observed. Arduino managed multiple components at once, with data handling and display efficiently. The reliability of the system makes it possible to operate for any application.

This project can be applied in smart homes, industrial temperature sensing, and farms. It may also be complemented with WiFi connectivity to enable remote monitoring through cloud platforms or mobile apps. Other features such as a buzzer will help give audio warnings for hazardous conditions.



Fig. 4: Temperature display in the LCD

In Briefly, the temperature monitoring system based on IoT performed consistently well to record and display temperature values and alert with the help of an LED. Further refinement like the use of IoT can further make it applicable in real-world usage.

VII. CONCLUSION

In this paper, we present the IoT-based Low-Cost Healthcare Solution Temperature Monitoring System which can be seen as a cost effective and reliable health monitor that can provide solution for healthcare especially for underprivileged or rural region. The system involves the use of an Arduino Uno microcontroller, an I2C type LCD display, and an NTC type thermistor sensor that is used to measure and display the temperature of the body of an individual in real time. Constant monitoring allows you to detect illnesses like fever or infection in time, which is why an inexpensive solution is what you need, that eliminates the need for manual temperature measurement.

The system is also upgradable, as it can be into a full-blown remote monitoring system with wireless modules Wi-Fi or Bluetooth an entire remote monitoring system with Wi-Fi or Bluetooth wireless module. The system is also very easy to install only requiring basic technical skills meaning it can easily be utilized by non-technocrats in far away rural areas. Using wireless communication modules (Wi-Fi, Bluetooth), this system can be established as a fully remote monitoring system providing online data to the doctors. The tool has the ability to revolutionize health services for rural

populations by enabling low-cost continuous vital sign monitoring. It could be used at homes, hospitals and health camps to monitor the temperature of the patient along with sending alerts to the doctor or nurse regarding any possible health threat. In the long run, it can integrate a sensor for heart rate, oxygen in the blood and storage of data in the cloud, increasing the system's usefulness for general health monitoring. The benefits of an IoT-based solution with DHT11 & LCD module for health services in rural areas includes - better health, through early detection of health needs and prompt action.

VIII. FUTURE WORK

Temperature Monitoring System based on the Internet of Things is being set up and there are some features that may enhance it. First of all, the wireless communication modules such as Wi-Fi or Bluetooth may be included. This will support remote monitoring with real-time data and alerts on unusual temperature measurements being provided to healthcare professionals or caregivers. Incorporation of cloud storage using software like ThingSpeak or Google Firebase will offer easy retrieval of historical data, enabling healthcare professionals to monitor trends and diagnose potential health issues.

Additional inclusion of additional sensors to detect parameters like heart rate, blood pressure, or oxygen level will turn the system into a comprehensive health monitoring system. Design of a web interface or mobile application will facilitate easier access and allow easy monitoring and access of multiple patients. Last but not least, portability and battery life optimization will render the system portable without infrastructure power in remote locations, enhancing field applications and rural health scenarios.

REFERENCES

- [1] S. K. Kumar, R. Manimegalai, A. Rajeswari, R. Deekshita, M. Dhineshkumar and G. Manikandan, "A Literature Review: Performance Evaluation of Wearable system with Pill Dispenser Box for Post Covid Elderly Patients," *2021 3rd International Conference on Advances in Computing, Communication Control and Networking (ICAC3N)*, Greater Noida, India, 2021, pp. 2008-2014, doi: 10.1109/ICAC3N53548.2021.9725733.
- [2] Kalimuthukumar, S., T. Arunprasath, S. Raja Lakshmi, M. Pandiarajan, S. Gowtham Pandi, and C. Mahendran. "Rehabilitation System Using Virtual Gaming for Spinal Cord Injury (SCI) Patients to Activate Sensorimotor Areas." In *International Conference on Intelligent Electrical Systems and Industrial Automation*, pp. 93-99. Singapore: Springer Nature Singapore, 2024.
- [3] N. Muralikumar, N. Ramanan, M. P. Rajasekaran and S. Kalimuthukumar, "IoT-Based Heart Rate Monitoring System Designed For Pilots," *2023 International Conference on Energy, Materials and Communication Engineering (ICEMCE)*, Madurai, India, 2023, pp. 1-6, doi: 10.1109/ICEMCE57940.2023.10434255.
- [4] S. K. Kumar, M. Dhineshkumar, A. Rajeswari, R. Manimegalai, R. M. Kalaranjini and G. Manikandan, "Development of an IoT Based Domestic Monitoring and Alerting System for Paralyzed Patients," *2021 3rd International Conference on Advances in Computing, Communication Control and Networking (ICAC3N)*, Greater Noida, India, 2021, pp. 586-591, doi: 10.1109/ICAC3N53548.2021.9725551.
- [5] S. Kalimuthukumar, G. Vagesvari, P. Thenmozhi, T. GirijaPragathi and N. Vigneshwari, "Early- Detection of Parkinson's Disease by Patient Voice Modulation Analysis through MFCC Feature Extraction Technique," *2021 3rd International Conference on Advances in Computing, Communication Control and Networking (ICAC3N)*, Greater Noida, India, 2021, pp. 1045-1049, doi: 10.1109/ICAC3N53548.2021.9725374.
- [6] K. Sakthivel, P. Thavamani, R. Chandrasekar, M. s. m. Ismail, V. Govindaraj and S. sankaran, "A Literature Review: A Novel Human Vital Sign based Wheel Chair Cum Stretcher for Disabled Person," *2020 International Conference on Communication and Signal Processing (ICCSPP)*, Chennai, India, 2020, pp. 1494-1500, doi: 10.1109/ICCSPP48568.2020.9182260.
- [7] I. Amir Javadpour, Hamidreza Memarzadeh- Tehran, Fatemeh Saghafi, "A Temperature Monitoring System Incorporating an Array of Precision Wireless Thermometer" on 2015
- [8] Nor Alina Khairi, Asral Bahari Jambek, Teoh Wei Boon, Uda Hashim, "Design and Analysis of a Wireless Temperature Monitoring System" on 2013, RSM2013 Proc.
- [9] Gungur Bal, Nihat Daldal, "Design and Implementation based Temperature Measurement and Control System using Powerline Communications" on 2013 at 2013 Fourth International Conference.
- [10] Myeong-in Choi et al., "Design and implementation of IoT-based HVAC system for future zero energy building," *2017 IEEE International Conference on Pervasive Computing and Communications Workshops (PerCom Workshops)*, Kona, HI, 2017, pp. 605-610, doi: 10.1109/PERCOMW.2017.7917631.
- [11] R. K. Kodali, V. Jain and S. Karagwal, "IoT based smart greenhouse," *2016 IEEE Region 10 Humanitarian Technology Conference (R10-HTC)*, Agra, 2016, doi: 10.1109/R10-HTC.2016.7906846
- [12] Tacklim Lee, Seonki Jeon, Dongjun Kang, Lee Won Park and Sehyun Park, "Design and implementation of intelligent HVAC system based on IoT and Bigdata platform," *2017 IEEE International Conference on Consumer Electronics (ICCE)*, Las Vegas, NV, 2017, pp. 398-399, doi: 10.1109/ICCE.2017.7889369.